RK-A-ROOM · REV 3 · 05 SEP 2026 · TERRACE PLANT · 11 RACKS

CEA Room Floor Plan

A 20 × 12 ft room set out from the validated rack envelope, with the nutrient plant moved to the terrace and only a 1.02 m² service corner left indoors. Eleven racks, every rack coordinate fixed, every service routed, and the two constraints that decide the room's clock stated plainly.

ROOM	RACKS	BEDS	GROW AREA
6096 × 3658	**11**	**44 · 176 trays**	**29.0 m²**
FLOOR MULTIPLIER	MAX DEMAND		
1.30×	**5.7 kW**		

01 · SETTING OUT

What changed, and what it was worth

Rev 2 gave the room a 5.73 m² water plant bay and nine racks. Moving both reservoirs to the terrace leaves one service corner and buys back two racks.

```
room internal 6096 × 3658 mm = 22.3 m²
rack installed envelope 1456 × 690 × 1960 mm (integrated model Rev 4, interference-clean)

across the 3658 mm depth:
 3 rows × 690 = 2070 → 2 aisles of 794 mm
 4 rows would need 4×690 + 3×700 = 4860 → does not fit. Three rows is the maximum

along the 6096 mm length:
 4 racks at 1476 pitch = 5904, ending at 5984, 112 mm to the east wall
 row A gives up its east bay for the service corner
 11 racks + a 1476 × 690 service corner

grow area 11 × 4 × (1.176 × 0.560) = 29.0 m²
multiplier 29.0 / 22.3 = 1.30× (was 1.06× with the plant indoors)
```

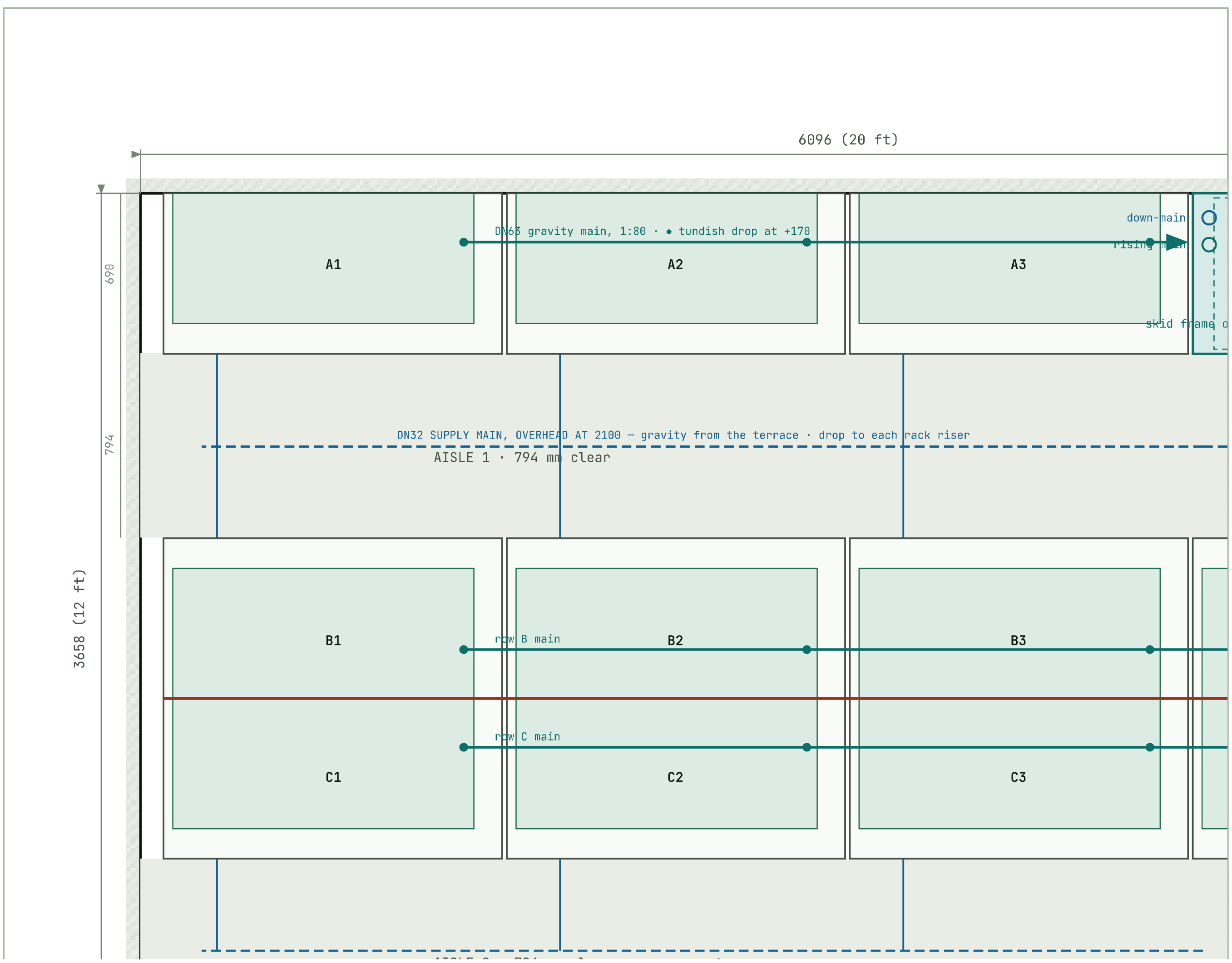
Rows B and C sit back to back at Y = 2174, bolted to each other through the wall-anchor brackets — that pair is self-stabilising, which removes eight of the eleven racks' dependence on the building fabric. Row A anchors to the north wall.

> **Set the room out for 690 mm even though Phase 1 is only 648 mm deep.** With the plenum deferred (§6) the deepest item on a rack is the supply riser at Y 643, so a Phase-1 rack is 648 mm deep. Do *not* use that to tighten the rows. Setting out at the final 690 mm means the plenum retrofit in month two or three is a bolt-on, not a room re-plan — and 42 mm × 3 rows is 126 mm of aisle you will never notice.

---

02 · PLAN

General arrangement



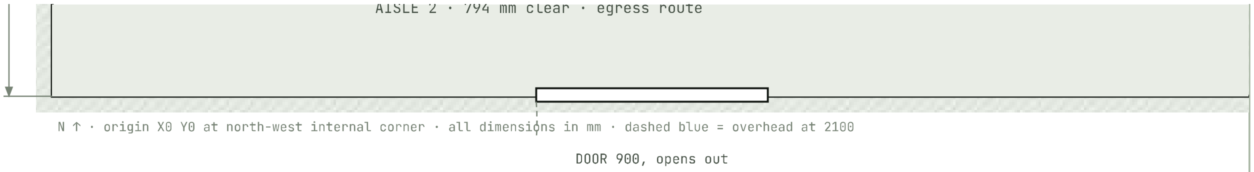


FIG 1 — REV 3 GENERAL ARRANGEMENT. BLUE DASHED IS THE GRAVITY SUPPLY MAIN, OVERHEAD AT 2100 MM WITH A DROP TO EACH RACK. TEAL IS DN63 GRAVITY DRAINAGE, RUN INSIDE THE RACK FOOTPRINTS AND COLLECTED IN THE 112 MM STRIP AGAINST THE EAST WALL.

Rack setting-out

RACK	X LEFT	Y FRONT	FACES	SUPPLY RISER X	TUNDISH X	TUNDISH Y	ANC
A1	100	690	South, Aisle 1	330	1451	210	Nor × M
A2	1576	690	South, Aisle 1	1806	2927	210	Nor × M
A3	3052	690	South, Aisle 1	3282	4403	210	Nor × M
B1	100	1484	North, Aisle 1	330	1451	1964	C1, r rear
B2	1576	1484	North, Aisle 1	1806	2927	1964	C2, rear
B3	3052	1484	North, Aisle 1	3282	4403	1964	C3, rear
B4	4528	1484	North, Aisle 1	4758	5879	1964	C4, rear
C1	100	2864	South, Aisle 2	330	1451	2384	B1, r rear
C2	1576	2864	South, Aisle 2	1806	2927	2384	B2, rear
C3	3052	2864	South, Aisle 2	3282	4403	2384	B3, rear

C4	4528	2864	South, Aisle 2	4758	5879	2384	B4, rear
Service corner	4528	0	**1476 × 690 · TK-02 catch trough with the recovery skid on a frame abov panel on the east wall**				

03 · WATER

Down by gravity overhead, back up by pump in one corner

Supply is overhead and dry; drainage is at floor level and inside the rack footprints. Neither crosses a walking surface.

Supply — gravity, overhead at 2100 mm

The DN32 down-main enters the room through the slab in the service corner, runs at 2100 mm above both aisles, and drops to each rack's supply riser at the rack's left end. Overhead because there is no pressure to lose — 2.66 m of static head against 0.33 m of loss — and because a pipe at 2100 mm is out of everyone's way. The master valve MV-01 on the terrace is the isolation point for the whole room; each rack keeps its own ball valve and union.

Drainage — gravity, in the rack footprints

```
Fall budget, worst case — rack C1 to TK-02
along row C, 1451 → 6040 = 4.59 m
north up the east strip, 2384 → 345 = 2.04 m
total 6.63 m at 1:80 → 83 mm fall
tundish outlet 170 - 83 = trough inlet at 87 mm AFFL
trough working volume 1.400 × 0.650 × 0.087 = 79 L against a 46 L rack batch — pass

Peak hydraulic load one rack draining, one tier at a time = 30.3 L/min
DN63 at 1:80 half-full ≈ 132 L/min → utilisation 23% — pass
```

> **The east-wall collector runs in a 112 mm strip, and that strip is not a walking route.** Racks in rows B and C end at X 5984; the wall is at 6096. The north-south collector sits in that gap, crossing the ends of both aisles. It is a dead end past the last rack, but it must be covered with a removable GRP channel cover and marked — this is the one place in the room where a pipe is

> anywhere near a foot.

Water balance — 44 beds, three cycles a day

STREAM	L / CYCLE	L / DAY	M³ / YR
Gross flood delivered	638	1,914	632
Consumed — ET, harvest, substrate	29.6	88.9	29.3
Returned to trough	608	1,825	602
Blowdown at 8%	48.7	146	48.2
Make-up required	**78.3**	**235**	**77.6**

Recovery ratio **87.7%**; annual discharge 48.2 m^3 against 632 m^3 drain-to-waste — a saving of **554 m^3**. Full architecture, equipment and the reuse gate are in RK-A-WRS Rev 3.

> **The drain side sets the room's clock, not the fill side.** Gravity can flood all four tiers of a rack at once — 2 minutes — but recovery is one rack per batch, because the reuse gate is decided on a settled 46 L batch in a 79 L trough. Drain 2.7 min, settle 0.5, test 3, transfer 2 ≈ **8 minutes per rack, 88 minutes for eleven, three times a day**. That is 4.4 hours a day of drain-and-recover, and it is the number to design the schedule around.

---

04 · VENTILATION

Standalone EC fans now, plenum in month two or three

The plenum is a real piece of ducting design and it deserves its own iteration. Phase 1 runs the same EC fans on grid-mounted brackets; the plenum retrofits onto the same 50 mm hole grid without touching anything else.

	PHASE 1 — STANDALONE FANS	PHASE 2 — DUCTED PLENUM
Hardware per tier	One 120 × 120 × 38 EC fan on a grid-mounted bracket at the rear-right of the bed	Folded 0.8 mm GI plenum, 1040 × 120 × 170, perforated face, same fan in the end cap

Air delivered	178 m³/h	178 m³/h — unchanged
Velocity uniformity	Point source. Expect a coefficient of variation near **33%** across the bed	Distributed along the full 1176 mm. CFD literature reports about **10%**
Installed rack depth	648 mm	690 mm
Cost per rack	₹2,880	₹5,520 · retrofit delta ₹2,640
Retrofit path	The plenum bolts to the same grid holes the fan bracket uses. No structural, electrical or plumbing change; the fan and its 24 V + PWM + tacho cable move across unaltered	

```
deferred now, 11 racks × ₹2,640 = ₹29,040
what it buys later: uniformity CV 33% → 10%, and tier-to-tier consistency you can
actually control

Commission Phase 1 with the measurement that decides Phase 2:
hot-wire anemometer, 5 points per tier, at canopy height, fans at design PWM
 CV ≤ 20% → the plenum is an optimisation
 CV > 20% → the plenum is a fix, and the data tells you where to put the
perforations
```

> **This is the right sequencing, and the measurement is what makes it right.** Designing a plenum before you have measured the bed it serves means guessing at the perforation pattern. Running Phase 1 with a bare fan and a five-point traverse per tier gives you a real velocity map to design against — so the month-two plenum is drawn from data rather than from a rule of thumb. Budget the anemometer (₹9,000–14,000 for a usable hot-wire unit); it is the cheapest instrument in this project and it closes two open items.

---

05 · POWER

Three phases, fourteen circuits

CIRCUIT	LOAD	W	PHASE	PROTECTION	CABLE
C1–C4	Racks A1, A3, B1	1,264	L1	6 A Type C + 30 mA Type	3C × 2.5

C1–C4	Racks A1–A3, B1	1,264	L1	6 A Type C + 30 mA Type A RCBO each	3C × 2.5 mm² Cu
C5–C8	Racks B2–B4, C1	1,264	L2	6 A Type C + 30 mA Type A RCBO each	3C × 2.5 mm² Cu
C9–C11	Racks C2–C4	948	L3	6 A Type C + 30 mA Type A RCBO each	3C × 2.5 mm² Cu
C12	HVAC, 2.0 TR inverter, **three-phase**	1,750	L1 L2 L3	16 A TP Type C + 30 mA Type A	4C × 2.5 mm² Cu
C13	Dehumidifier, 50 L/day	700	L1	10 A Type C + 30 mA Type A RCBO	3C × 1.5 mm² Cu
C14	Water plant — P-02 A/B, UV-01, DOS-01, MV-01	820	L3	16 A Type C + 30 mA Type A; 10 mA on UV-01	3C × 2.5 mm² Cu
C15	Room lighting, sockets, controller on UPS	200	L2	6 A Type B + 30 mA Type A RCBO	3C × 1.5 mm² Cu
Connected		6,946	**Maximum demand at 0.82 diversity ≈ 5.7 kW · 415 V TPN, 32 A main**		

```
phase currents at 230 V, with a three-phase HVAC sharing 583 W per phase:
 L1 = (1264 + 700 + 583) / 230 = 11.1 A
 L2 = (1264 + 200 + 583) / 230 = 8.9 A
 L3 = (948 + 820 + 583) / 230 = 10.2 A
imbalance = (11.1 - 8.9) / 11.1 = 20% — above the 15% target
eleven racks will not divide by three; the residue is trimmed at commissioning by moving
the dehumidifier or the water plant between phases once running currents are measured
```

> **Specify the HVAC as three-phase.** With eleven racks the rack load already divides 4/4/3, so a single-phase 2 TR unit lands 1.75 kW on one leg and pushes the imbalance past 35%. A three-phase unit shares it and brings the room inside 20% without any other change. Three-phase 2 TR inverter units are readily available in India and the price difference is small against the cable and the trouble it saves.

- **Type A RCBOs, not Type AC** — LED drivers and inverter compressors produce pulsating DC residual currents that a Type AC device can be blinded by. Not the default on most Indian consumer units; state it on the order.
- **Surge:** Type 2 SPD, 10 kA, at the main panel, 3 m maximum lead to the earth bar.

- **Earthing** per IS 3043 — dedicated earth pit, 6 mm² bonding to every rack frame, continuity ≤ 1 Ω measured and recorded per rack.
- **Emergency stop:** one latching mushroom head per aisle. Drops all fill solenoids, MV-01 on the terrace and all pumps; drain solenoids de-energise open so every bed empties.
- **Terrace circuit:** MV-01 and the TK-01 instrument block run on a spur from C14 with its own 10 mA RCBO, in UV-stable conduit. Everything on a roof is an outdoor installation.

---

06 · CLIMATE

The latent load is the design driver

```
sensible = rack electrical 3,476 W + in-room pump average 300 W = 3.78 kW
latent = ET 43.5 L/day = 1.81 kg/h × 2450 kJ/kg = 1.23 kW
total = 5.01 kW → 1.42 TR
sensible heat ratio = 3.78 / 5.01 = 0.75

specify 2.0 TR three-phase inverter (1.5 TR would run at 95% duty with no reserve)
plus a 50 L/day dehumidifier for the dark period, when sensible load collapses
but transpiration does not stop with it
```

The racks mix the room themselves: 11 × 4 × 178 m³/h is **7,832 m³/h** in a 60.2 m³ room, about 130 air changes an hour. No destratification fan — it would fight the tier fans rather than help them.

PARAMETER	SETPOINT	CONTROL	SENSOR AND POSITION
Air temperature	20–24 °C	HVAC	Room T/RH at 1.5 m, mid-aisle, clear of any fan discharge
Relative humidity	55–70%	Dehumidifier, 5% hysteresis	Same node
VPD	0.6–1.0 kPa	Computed from T and RH — never sensed	Controller
Canopy air velocity	0.2–0.5 m/s	Fan PWM per tier	Hot-wire anemometer, 5 points per tier, commissioning and quarterly
CO_2 enrichment	1000–1500 µmol/mol	Solenoid, PID on room sensor	NDIR at 1.5 m, mid-aisle

CO_2 safety	Alarm at 5000 ppm	Independent monitor, fail-closed solenoid	Second NDIR at 300 mm above floor
Solution temperature	≤ 26 °C, inhibit at 28	MV-01 interlock; chiller if Coimbatore data demands it	TE-01 in TK-01, on the terrace
Fresh air	0.3–0.5 ACH	Filtered make-up, G4 + F7	Interlocked to CO_2 demand

> **The terrace solves the CO_2 siting problem for free.** A 22 kg cylinder released into 60.2 m^3 produces roughly 12 m^3 of gas — about 20% of the room volume, and lethal well before anyone notices. With the plant already on the roof, **the cylinder goes there too**, with only the regulated line entering the room. The low-level safety monitor and its fail-closed solenoid are still required — they cover a failure in the line, which is the remaining case.

---

07 · SENSING AND CONTROL

Fifty-four sensors, allocated by level

LEVEL	SENSORS	COUNT	NODE
RACK	Air T/RH at tier 3 canopy · leak puck at the drain-header base · header high-level switch LSH-04	33	11 rack controllers
TERRACE	TK-01 EC · pH · temperature · level LT-02 · down-main flow totaliser	5	Terrace I/O node
RECOVERY	Trough EC · pH · temperature · floats ×3 · filter ΔP · UV intensity · flow switch	9	Water skid node
ROOM	CO_2 enrichment · CO_2 safety at low level · T/RH supply · T/RH return · smoke · floor flood	6	Room I/O node
PORTABLE	PAR quantum sensor · hot-wire anemometer	2	Manual
Total fixed	**Across 14 bus nodes**	**53**	**RS485 multi-drop, 120 Ω both ends**

Three control layers, each surviving the loss of the one above it: **rack controllers** own the tier fill/dwell/drain sequence, fan PWM and the leak and header interlocks, and hold a last-knownfill, dwell, drain sequence, fan PWM and the leak and header interlocks, and hold a last-known-good schedule; the **room controller** owns the drain token, the eight-condition reuse gate, dosing, HVAC and CO_2 setpoints, alarms and the local log; the **cloud** owns remote view and history and nothing the room depends on.

> **The terrace node is a real addition, and the bus has to reach it.** RS485 to the roof means a cable through the slab in the service corner, in UV-stable conduit, with the terrace node as one physical end of the chain and its 120 Ω terminator there. Do not spur it off the middle of the run — that is the classic way to make a bus that works on the bench and fails in the rain.

---

08 · CAPEX

What the room costs